Solinas primes and generator cohorts

2026-10-12 · Measured descriptive study · cohorts-20261012

Measured: None of the 20 corrected intervals excludes zero. These data did not resolve a mean difference in the prespecified features.

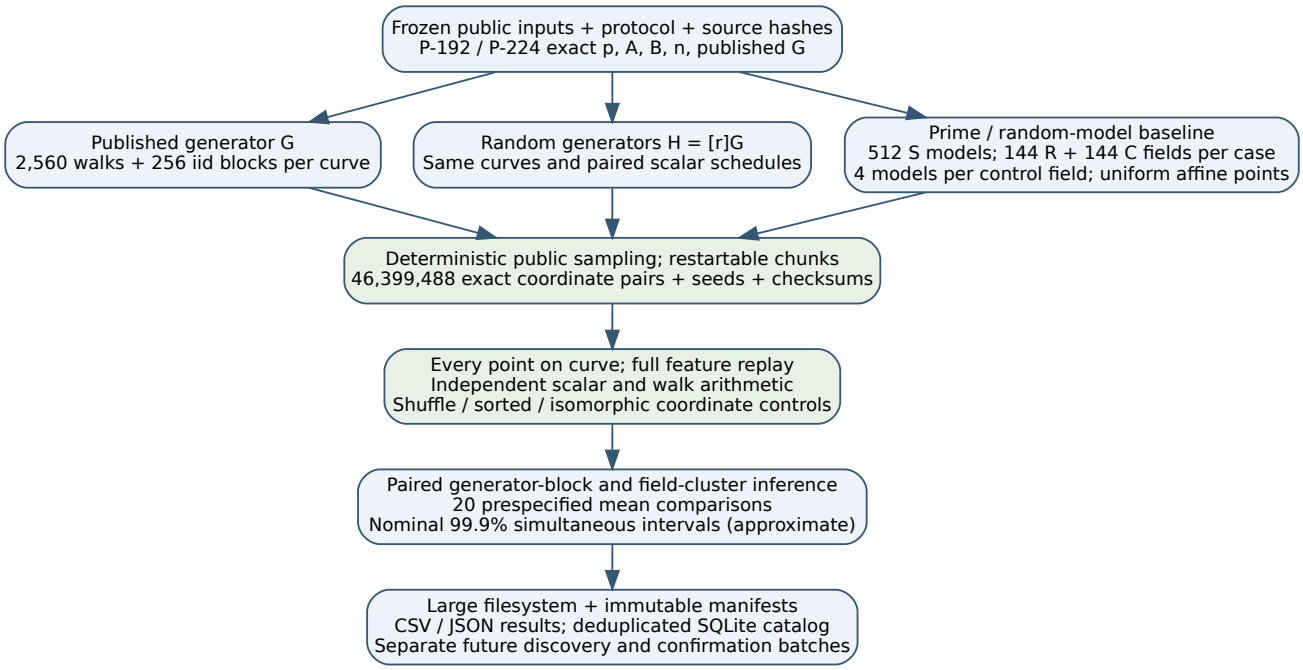
The prime, the curve, the generator

The prime defines residues modulo p. A and B define $y^2=x^3+Ax+B$. A generator defines an ordering of its finite group. P192 uses 2^192−2^64−1; P224 uses 2^224−2^96+1. The exact standard curves have prime group orders and cofactor 1.

Derived: H=rG, with r nonzero modulo prime n, permutes the nonzero multiples. Uniform kG and kH therefore have exactly the same point distribution. Ordered walks Q, Q+G, Q+2G versus Q, Q+H, Q+2H can expose ordering features. Consecutive points are dependent. Generator comparisons hold each standard curve fixed.

Fixed sample and controls

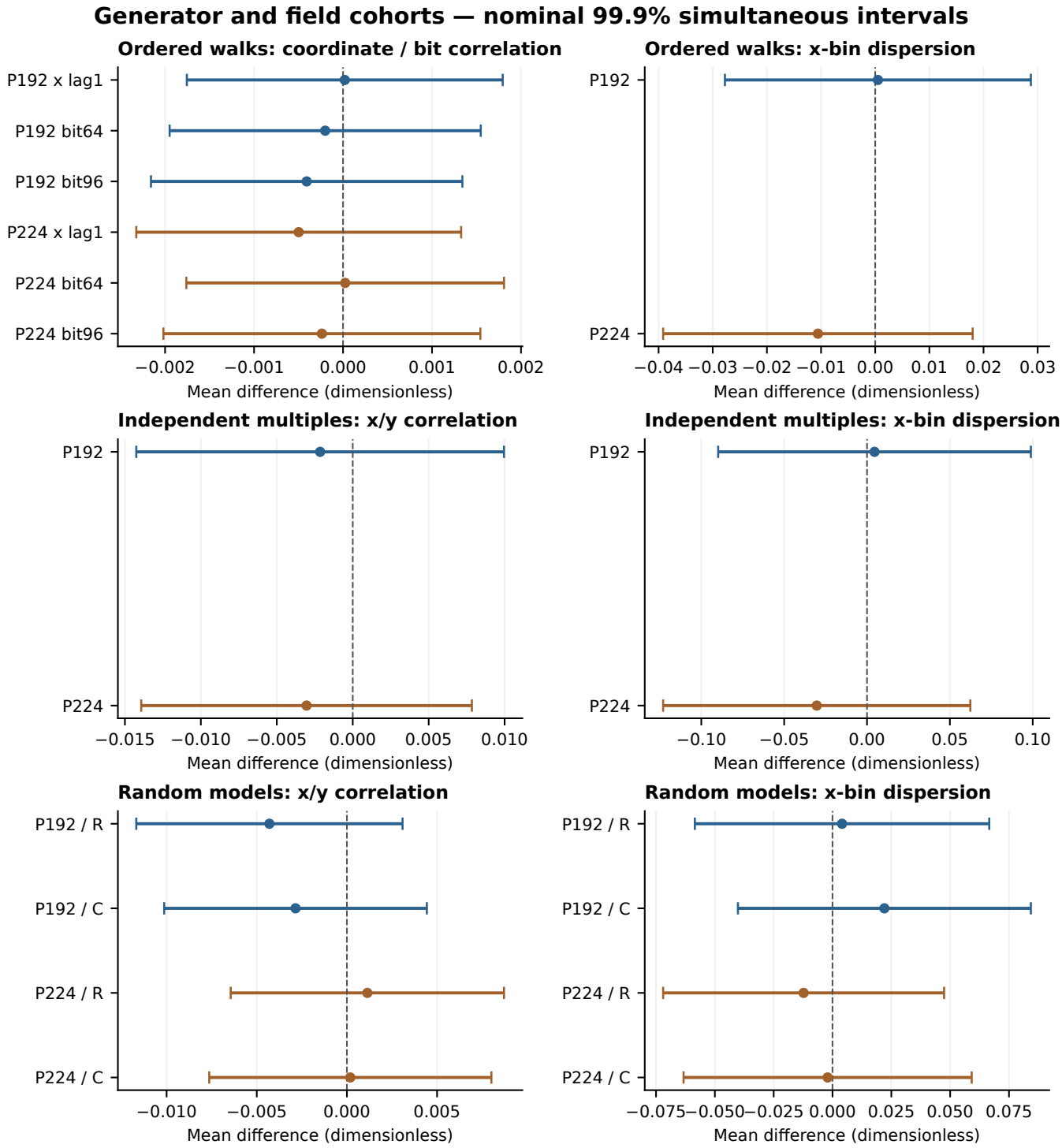
Each generator cohort on each standard curve: 2,560 random-start walks ×4,096 points (10,485,760) and 256 iid-multiple blocks ×1,024 (262,144). Random generators are fresh per block; published generators use identical public scalar schedules. Each case also uses 512 random models on fixed S p; 144 R and 144 C fields, four models/field, 1,024 uniform affine points/model. C matches p mod 12.



20 primary mean tests, nominal 99.9% simultaneous confidence via Bonferroni. Student/Welch approximations use generator blocks and control field means. The two fixed S primes are not a sample of all structured prime families.

Corrected comparison intervals

Published minus random G for walks/iid; fixed-S random models minus R/C field means for the field baseline. Correlations and bin dispersion are dimensionless. Each interval has nominal 99.995% confidence; simultaneous nominal 99.9% applies to these 20 tests in this batch. Sampling-distribution coverage is approximate.



Zero-compatible intervals bound tested mean features; they do not exclude every rare generator type, higher-order coordinate feature, or discovery from ML. No relative ECDLP resistance is measured. Official G is one fixed instance; many independent starts do not turn it into many independently chosen Gs.

Values and software checks

Case/population	Metric	Difference	Corrected interval	Units
P192/walk	x_lag1	1.9446e-05	(-0.0017556, 0.0017945)	2560/2560
P192/walk	bit64_lag1	−0.00020067	(-0.0019492, 0.0015479)	2560/2560
P192/walk	bit96_lag1	−0.00040879	(-0.0021588, 0.0013412)	2560/2560
P192/walk	x_bin_dispersion	0.00049557	(-0.027751, 0.028742)	2560/2560
P192/iid	x_bin_dispersion	0.0045363	(-0.089816, 0.098889)	256/256
P192/iid	xy_correlation	−0.002146	(-0.014252, 0.00996)	256/256
P192/field R	x_bin_dispersion	0.0040563	(-0.058508, 0.06662)	512/144
P192/field C	x_bin_dispersion	0.022054	(-0.040193, 0.084302)	512/144
P192/field R	xy_correlation	−0.0042983	(-0.011682, 0.0030855)	512/144
P192/field C	xy_correlation	−0.0028499	(-0.010134, 0.0044338)	512/144
P224/walk	x_lag1	−0.00049816	(-0.0023239, 0.0013275)	2560/2560
P224/walk	bit64_lag1	2.4061e-05	(-0.001761, 0.0018091)	2560/2560
P224/walk	bit96_lag1	−0.00023775	(-0.0020189, 0.0015434)	2560/2560
P224/walk	x_bin_dispersion	−0.010583	(-0.039162, 0.017996)	2560/2560
P224/iid	x_bin_dispersion	−0.030336	(-0.12304, 0.062366)	256/256
P224/iid	xy_correlation	−0.0030399	(-0.013925, 0.0078449)	256/256
P224/field R	x_bin_dispersion	−0.012278	(-0.071982, 0.047426)	512/144
P224/field C	x_bin_dispersion	−0.0020713	(-0.063298, 0.059155)	512/144
P224/field R	xy_correlation	0.0011342	(-0.0064444, 0.0087129)	512/144
P224/field C	xy_correlation	0.00018856	(-0.0076359, 0.008013)	512/144

Verified: 46,399,488 points on their exact curves; 14,592 SHA-256/feature replays; 163,840 independent walk transitions; 768 scalar arithmetic checks; 524,288 isomorphic transports. Separate Python affine arithmetic checks OpenSSL outputs. This is same-session software verification.

The first 32 blocks/cohort/case also test shuffle, sorted positive dependence, and u=2 isomorphic coordinate transport on exactly the same points. Sorted sequences are deliberately synthetic. These diagnostics do not enter primary multiplicity or create a security claim. See the Markdown report for values.

Per-cohort walk units =2,560; iid units =256. Field baseline uses 512 fixed-prime coefficient draws versus 144 independent field means per control arm. The data reuse 578 prime fields. The published generator exists only on its standard curve; the field baseline does not identify a universal prime-by-G interaction.

Sampling size and a reusable runner

The **sampling fraction** is sampled points / population size. A fixed fraction of a 192-bit field is infeasible. Ten million / P192 p is about 1.6×10^−51. **Power analysis** uses effect size, variance, confidence and independent units. **Rare-event detection** uses event probability q: independent draws needed for 99.9% chance of at least one recognizable event are ceil(log(.001)/log(1−q)).

Event probability q	Independent draws
1%	688
0.1%	6,905
0.01%	69,075
1 in a million	6,907,752

For rare fields count independent fields; for rare generators count generator draws. Walk points do not supply independent event trials. The finite-population correction is negligible at these fractions and applies to simple sampling without replacement, not to walks. Variance-based detectable effects appear in the Markdown comparison table; they do not bound all possible patterns.

Run batches without an agent

Linux, Python 3.10+, C compiler and OpenSSL development headers. A standalone package bundles public inputs. Each batch freezes source, seed and configuration, then writes restartable exact coordinate chunks, SHA-256 manifests and features.

```
bash bootstrap.sh
bash run_batch.sh /large-filesystem/ec-study/batch-0001 batch-0001 3
.venv/bin/python catalog.py --root /large-filesystem/ec-study --output /large-filesystem/ec-study/catalog.sqlite
```

The same batch command resumes; a new seed gets a new output directory. A deduplicated SQLite index retains field/model/generator identities for analysis and ML. Default raw bytes: 2,412,773,376 (2.41 GB). Measured wall time: 577.7s, job CPU sum 1654.7s, three workers.

New seeds add conditional point/start/G replicates; they add no new primes. Use independent confirmation seeds and field/curve-aware ML splits. Repeated batch inference needs a prospectively declared pooled or sequential policy.

Sources and identities

Frozen source: `18e4cca389b76f208b8fec37c1f0c73eb24a1f1c`. Full exact standard EC1 identities and parameter preimages are in inputs.json and report.md; all random-G identities retain the changed generator. Public baseline inputs: EXP-ECDLP-5184f7 / RUN-ECDLP-d0e449. Receipt, analysis, verification, lossless record shards, source/PDF, graphs and data bundles are linked from report.md. No canonical frontier edge changes; no hypothesis promotion or security claim.